MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, TTL, LOW POWER,
EXCLUSIVE-OR GATES, MONOLITHIC SILICON

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, TTL, low power, quadruple, 2-input exclusive-OR gate microcircuits. Three product assurance classes and a choice of case outline/lead finish are provided and are reflected in the complete part number.

1.2 Part number. The complete part number shall be as shown in the following example:

M38510	/026	01	B	A	C
Military designator	Detail specification	Device type (1.2.1)	Device class (1.2.2)	Case outline (1.2.3)	Lead finish (3.3)

1.2.1 Device type. The device type shall be as follows:

Device type	Circuit
01	Quadruple, 2-input exclusive-OR gate

1.2.2 Device class. Device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

Letter	Case outline, MIL-M-38510, appendix C
A	F-1 (14-pin, 1/4" x 1/4", flat pack)
B	F-3 (14-pin, 1/8" x 1/4", flat pack)
C	D-1 (14-pin, 1/4" x 3/4", dual-in-line)
D	F-2 (14-pin, 1/4" x 3/8", flat pack)

1.3 Absolute maximum ratings:

Supply voltage range	0 Vdc to 8.0 Vdc
Input voltage range	0 Vdc to 5.5 Vdc
Storage temperature range	-65° C to 150° C
Maximum power dissipation, P_D 1/	9 mWdc per gate
Lead temperature (soldering 10 seconds)	300° C
Thermal resistance, junction to case	θ_{JC} = { 0.15° C/mW for flat pack; 0.08° C/mW for dual-in-line pack
Junction temperature	T_J = 175° C

1.4 Recommended operating conditions:

Supply voltage	4.5 Vdc minimum to 5.5 Vdc maximum
Minimum high level input voltage	2.0 Vdc
Maximum low level input voltage	0.7 Vdc
Normalized fanout (each output) 2/	10 maximum
Ambient operating temperature range	-55° C to 125° C

1/ Must withstand the added P_D due to short circuit condition (e.g., I_{OS}) at one output for 5 seconds duration.
2/ Device will fanout in both high and low levels to the specified number of inputs of the same device type as that being tested.

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2. APPLICABLE DOCUMENT

2.1 The following document, of the issue in effect on date of invitation for bids or request for proposal, forms a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Detail specifications. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein. In the event of conflict between MIL-M-38510 and this detail specification, this detail specification shall govern.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510, and herein.

3.2.1 Case outline.

3.2.1.1 Case outline A. Case outline A shall be in accordance with F-1 of Appendix C to MIL-M-38510.

3.2.1.2 Case outline B. Case outline B shall be in accordance with F-3 of Appendix C to MIL-M-38510.

3.2.1.3 Case outline C. Case outline C shall be in accordance with D-1 of Appendix C to MIL-M-38510.

3.2.1.4 Case outline D. Case outline D shall be in accordance with F-2 of Appendix C to MIL-M-38510.

3.2.2 Logic diagram and terminal connections. The logic diagram and terminal connections shall be as specified on figure 1.

3.2.3 Truth tables and logic equations. The truth tables and logic equations shall be as specified on figure 2.

3.2.4 Schematic circuit. The schematic circuit shall be as specified on figure 3.

3.3 Lead material and finish. Lead material and finish shall be in accordance with MIL-M-38510.

3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I, and apply over the full recommended ambient operating temperature range, unless otherwise specified.

3.5 Rebonding. Rebonding shall be in accordance with MIL-M-38510.

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TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions	Limits		Units
			Min	Max	
High-level output voltage	V_{OH}	V_{CC} = 4.5V, V_{IN} = 2.0V V_{IN} = 0.7V, I_{OH} = -100 μA 1/	2.4	---	Volts
Low-level output voltage	V_{OL}	V_{CC} = 4.5V, V_{IN} = 2.0V (all) or V_{IN} = 0.7V (all) I_{OL} = 2 mA 1/	---	0.3	Volts
High-level input current	I_{IH1}	V_{CC} = 5.5V, V_{IN} = 2.4V 2/	---	20	μA
High-level input current	I_{IH2}	V_{CC} = 5.5V, V_{IN} = 5.5V 2/	---	80	μA
Low-level input current	I_{IL}	V_{CC} = 5.5V, V_{IN} = 0.3V 1/	---	-0.36	mA
Short-circuit output current	I_{OS}	V_{CC} = 5.5V, V_{IN} = 5.5V V_{IN} = 0.0V, 2/ 3/	-3	-15	mA
High-level supply current per gate	I_{CCH}	V_{CC} = 5.5V, V_{IN} = 5.5V V_{IN} = GND 2/	---	1.1	mA
Low-level supply current per gate	I_{CCL_1}	V_{CC} = 5.5V V_{IN} = 0.0V (all inputs)	---	1.67	mA
Low-level supply current per gate	I_{CCL_2}	V_{CC} = 5.5V, V_{IN} = 5.5V (all inputs)	---	1.67	mA
Propagation delay time high-to-low (other input low)	t_{PHL_1}	V_{CC} = 5.0V R_L = 4 KΩ C_L = 50 pF	7	165	ns
Propagation delay time high-to-low (other input high)	t_{PHL_2}		7	80	ns
Propagation delay time low-to-high (other input low)	t_{PLH_1}		7	180	ns
Propagation delay time low-to-high (other input high)	t_{PLH_2}		7	110	ns

1/ All unspecified inputs at 5.5 volts.
2/ All unspecified inputs at 0.0 volts.
3/ Not more than one output should be shorted at one time.

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3.6 Electrical test requirements. Electrical test requirements shall be as specified in table III for the applicable device class. The subgroups of table III which constitute the minimum electrical test requirements for screening, qualification, and quality conformance by device class are specified in table II.

TABLE II. Electrical test requirements.

MIL-STD-883 test requirement	Subgroups (see table III) Class A devices	Class B devices	Class C devices
Interim electrical parameters (pre burn-in) (method 5004)	1	1	None
Final electrical test parameters (method 5004)	1*, 2, 3, 9	1*, 2, 3, 9	1
Group A test requirements (method 5005)	1, 2, 3, 9, 10, 11	1, 2, 3, 9	1, 2, 3, 9
Groups B and C end point electrical parameters (method 5005)	1, 2, 3	1, 2, 3	1
Additional electrical subgroups for group C periodic inspections	None	10, 11	None

* PDA applies to subgroup 1 (see 4.3(h)).

3.7 Marking. Marking shall be in accordance with MIL-M-38510 and 1.2. At the option of the manufacturer, the following marking may be omitted from the body of the microcircuit, but shall be retained on the initial container:

(a) Country of origin.
(b) Manufacturer's identification.

4. PRODUCT ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and Method 5005 of MIL-STD-883, except as modified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified herein for groups A, B, and C inspections (see 4.4.1, 4.4.2, and 4.4.3). After qualification of one or more electrically and structurally similar types with a single lead finish, other lead finishes of the same case outline may be qualified by submitting a single type in the qualified case outline to the group B, subgroup 3 test and the group C, subgroups 1, 3, and 4 tests.

4.3 Screening. Screening shall be in accordance with Method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

(a) Test samples for the group B bond strength test specified in Method 5005 of MIL-STD-883 may, at the manufacturer's option, be randomly selected immediately following the internal visual (precap) inspection and prior to sealing (see 4.4.2(b)).
(b) Temperature cycling (Method 1010 of MIL-STD-883).
    (1) Omit seal test as post-test measurement.
(c) Thermal shock (Method 1011 of MIL-STD-883), when substituted for temperature cycling.
    (1) Omit seal test as post-test measurement.

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(d) Burn-in test (Method 1015 of MIL-STD-883).
  (1) Test condition D or E, using the circuit shown on figure 4, or equivalent, subject to approval of qualifying activity.
  (2) T_A = 125° C minimum.
(e) Reverse bias burn-in and interim electrical test in accordance with 3.1.10 of Method 5004 of MIL-STD-883 may be omitted.
(f) Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.
(g) External visual inspection shall not include measurement of case and lead dimensions.
(h) Percent defective allowable (PDA) - The PDA is specified as 5 percent for class A devices and 10 percent for class B devices based on failures from group A, subgroup 1 test after cooldown as final electrical test in accordance with method 5004 of MIL-STD-883, and with no intervening electrical measurements. If interim electrical parameter tests are performed prior to burn-in, failures resulting from pre burn-in screening may be excluded from the PDA. If interim electrical parameter tests prior to burn-in are omitted, then all screening failures shall be included in the PDA. The verified failures of Group A, subgroup 1 after burn-in divided by the total number of devices submitted for burn-in in that lot shall be used to determine the percent defective for that lot, and the lot shall be accepted or rejected based on the PDA for the applicable device class.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510.

4.4.1 Group A inspection. Group A inspection shall consist of the test subgroups and LTPD values shown in table I of Method 5005 of MIL-STD-883 and as follows:

(a) Tests shall be as specified in table II.
(b) Subgroups 4, 5, 6, 7, and 8 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall consist of the test subgroups and LTPD values shown in table II of Method 5005 of MIL-STD-883 and as follows:

(a) End point electrical parameters shall be as specified in table II.
(b) Bond strength test may be conducted on samples collected prior to sealing (see 4.3(a)).

4.4.3 Group C inspection. Group C inspection shall consist of the test subgroups and LTPD values shown in table III of Method 5005 of MIL-STD-883 and as follows:

(a) End point electrical parameters shall be as specified in table II.
(b) Subgroups 7 and 8 shall be added to the group C inspection requirements for class B devices and shall consist of the tests, conditions and limits specified for subgroups 10 and 11 of group A.
(c) Lead bend in only one direction is required for initial conditioning prior to moisture resistance and salt atmosphere tests.
(d) High temperature storage test (Method 1008 of MIL-STD-883) conditions:
  (1) Temperature: 150 ±10° C.
  (2) Duration: 1,000 hours, except as otherwise permitted by Appendix B of MIL-M-38510.
(e) Operating life test (Method 1005 of MIL-STD-883) conditions:
  (1) Test condition D or E, using the circuit shown on figure 4, or equivalent, subject to approval of qualifying activity.
  (2) T_A = 125° C, minimum.
  (3) Test duration: 1,000 hours, except as permitted by Appendix B of MIL-M-38510.
(f) Omit steady state reverse bias test.

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Device type 01

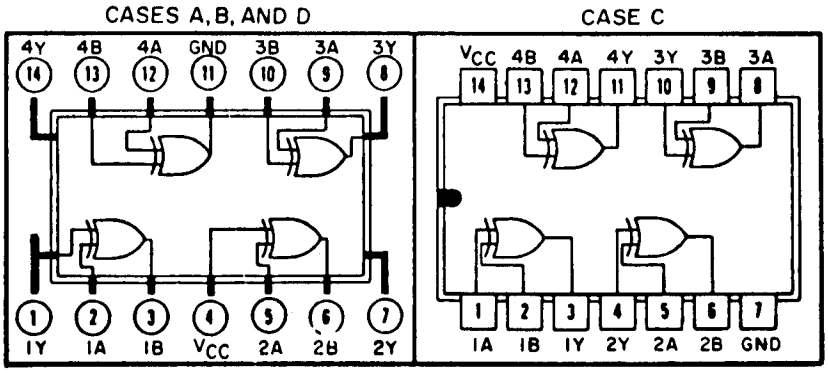


FIGURE 1. Logic diagram and terminal connections.

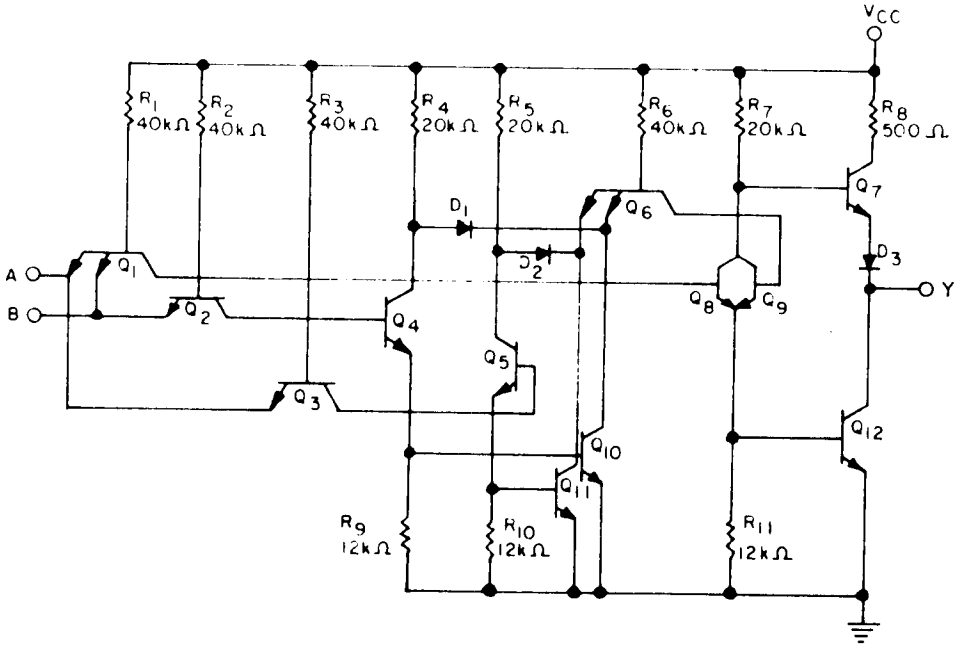
TRUTH TABLE (each gate)

INPUTS		OUTPUTS
A	B	Y
L	L	L
L	H	H
H	L	H
H	H	L

Positive logic: $Y = A \oplus B$

FIGURE 2. Truth table.

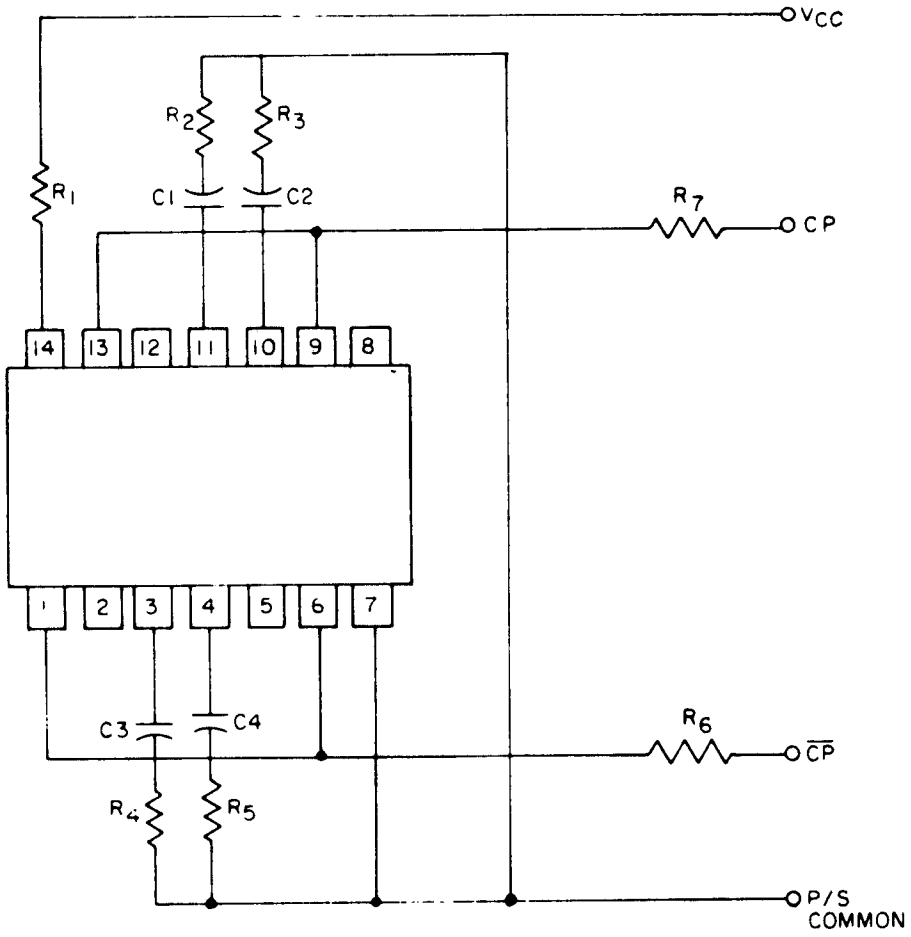
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NOTE: All resistance values shown are nominal.

FIGURE 3. Schematic circuit.

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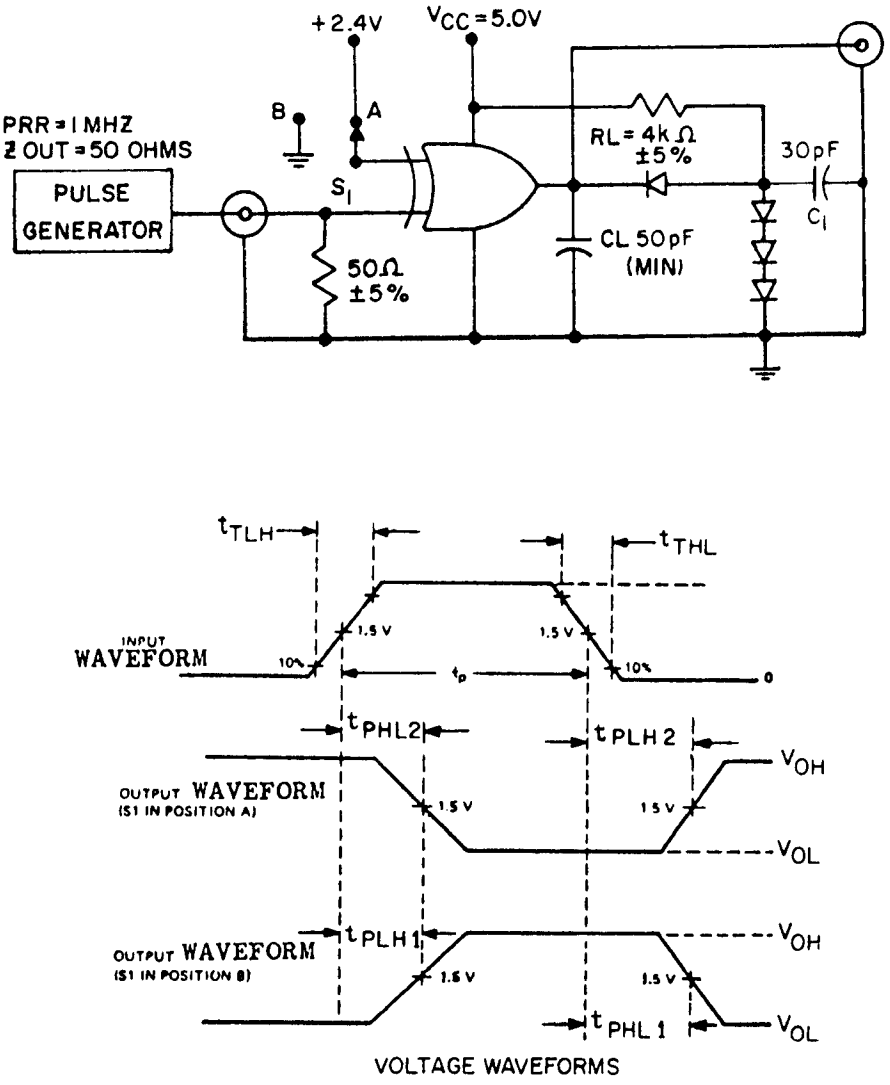


NOTES:
R1 = 26.1 Ω
R2 thru R5 = 976 Ω
R6, R7 = 26.1 Ω
C1 thru C4 = .0047 μF

NOTES:
1. Tolerance on resistors ±5%; on capacitors, ±10%.
2. V_{CC} and R_1 shall be such that 5.0 volts appears at the device terminal.
3. If parallel excitation is used, clock frequency shall be 100 KHz with minimum V_{IN} = 3V and minimum duty cycle of 50%.

FIGURE 4. Burn-in and life test circuit.

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NOTES:
1. The generator has the following characteristics: V_{gen} = 3V, $t_{PHL} = t_{PLH} \leq 60$ ns, t_p = 1.0 μs, PRR $\leq$ 500 KHz, $Z_{out} \approx 50\ \Omega$.
2. All diodes are 1N3064.
3. C_L includes probe and jig capacitance.
4. Each gate tested separately.

FIGURE 5. Switching time test circuit.

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TABLE III. Group A inspection for device type 01.
Terminal conditions (pins not designated are open)

Subgroup	Symbol	MIL-STD-883 method	Case A, B, D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Meas. terminal	Test limits		
			Case C	3	1	2	14	5	6	4	10	8	9	7	12	13	11		Min	Max	Unit
			Test No.	1Y	1A	1B	VCC	2A	2B	2Y	3Y	3A	3B	GND	4A	4B	4Y				
1 T_A=25° C	V_{OL}	3007	1	2 mA	0.7 v	0.7 v	4.5 v	5.5 v	5.5 v			5.5 v	5.5 v	GND	5.5 v	5.5 v		1Y		0.3	V
			2	2 mA	2.0 v	2.0 v		5.5 v	5.5 v									1Y			
			3		5.5 v	5.5 v		0.7 v	0.7 v	2 mA								2Y			
			4					2.0 v	2.0 v	2 mA								2Y			
			5					5.5 v	5.5 v		2 mA	0.7 v	0.7 v					3Y			
			6								2 mA	2.0 v	2.0 v					3Y			
			7									5.5 v	5.5 v		0.7 v	0.7 v	2 mA	4Y			
			8												2.0 v	2.0 v	2 mA	4Y			
	V_{OH}	3006	9	-100 μA	2.0 v	0.7 v									5.5 v	5.5 v		1Y	2.4		
			10	-100 μA	0.7 v	2.0 v												1Y			
			11		5.5 v	5.5 v		2.0 v	0.7 v	-100μA								2Y			
			12					0.7 v	2.0 v	-100μA								2Y			
			13					5.5 v	5.5 v		-100 μA	0.7 v	2.0 v					3Y			
			14								-100 μA	2.0 v	0.7 v					3Y			
			15									5.5 v	5.5 v		0.7 v	2.0 v	-100μA	4Y			
			16									5.5 v	5.5 v		2.0 v	0.7 v	-100μA	4Y			
	I_{IH1}	3010	17		2.4 v	GND	5.5 v	GND	GND			GND	GND		GND	GND		1A		20	μA
			18		GND	2.4 v		GND	GND									1B			
			19			GND		2.4 v	2.4 v			2.4 v						2A			
			20					GND	GND			GND	2.4 v		2.4 v			2B			
			21										GND		GND	2.4 v		3A			
			22															3B			
			23															4A			
			24															4B			
	I_{IH2}		25		5.5 v											GND		1A		80	
			26		GND	5.5 v												1B			
			27			GND		5.5 v	5.5 v									2A			
			28					GND	GND									2B			
			29									5.5 v						3A			
			30									GND	5.5 v					3B			
			31										GND		5.5 v	5.5 v		4A			
			32										GND		GND			4B			
	I_{IL}	3009	33		0.3 v	5.5 v		5.5 v	5.5 v			5.5 v	5.5 v		5.5 v			1A		-.36	mA
			34		5.5 v	0.3 v		5.5 v	5.5 v									1B			
			35			5.5 v		0.3 v	0.3 v									2A			
			36					5.5 v	5.5 v									2B			
			37									0.3 v						3A			
			38									5.5 v	0.3 v					3B			
			39										5.5 v		0.3 v			4A			
			40										5.5 v		5.5 v	0.3 v		4B			
	I_{OS}	3011	41	GND		GND		GND	GND			GND	GND					1Y	-3	-15	
			42		GND				5.5 v	GND		GND			GND	GND		2Y			
			43						GND		GND	5.5 v						3Y			
			44						GND			GND				5.5 v	GND	4Y			
	I_{CCH}	3005	45		5.5 v				5.5 v			5.5 v				5.5 v		VCC		4.4	
	I_{CCL1}		46		GND				GND			GND				GND		VCC		6.68	
	I_{CCL2}		47		5.5 v	5.5 v		5.5 v	5.5 v			5.5 v	5.5 v		5.5 v	5.5 v		VCC		6.68	
2	Same tests, terminal conditions and limits as subgroup 1, except T_A = 125° C.																				
3	Same tests, terminal conditions and limits as subgroup 1, except T_A = -55° C.																				

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TABLE III. Group A inspection for device type 01. - Continued
Terminal conditions (pins not designated are open)

Subgroup	Symbol	MIL-STD-883 method	Case A, B, D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Meas. terminal	Test limits		
			Case C	3	1	2	14	5	6	4	10	8	9	7	12	13	11				
			Test No.	1Y	1A	1B	VCC	2A	2B	2Y	3Y	3A	3B	GND	4A	4B	4Y		Min	Max	Unit
9 T_A=25° C	t_{PHL_1}	3003 (Fig. 5)	48 49 50 51	OUT	IN	0.7 v	5.0 v	IN	0.7 v	OUT	OUT	IN	0.7 v	GND	IN	0.7 v	OUT	1A to 1Y 2A to 2Y 3A to 3Y 4A to 4Y	7	105	ns
	t_{PHL_2}		52 53 54 55	OUT	IN	2.0 v		IN	2.0 v	OUT	OUT	IN	2.0 v		IN	2.0 v	OUT	1A to 1Y 2A to 2Y 3A to 3Y 4A to 4Y	7	50	
	t_{PLH_1}		56 57 58 59	OUT	IN	0.7 v		IN	0.7 v	OUT	OUT	IN	0.7 v		IN	0.7 v	OUT	1A to 1Y 2A to 2Y 3A to 3Y 4A to 4Y	7	115	
	t_{PLH_2}		60 61 62 63	OUT	IN	2.0 v		IN	2.0 v	OUT	OUT	IN	2.0 v		IN	2.0 v	OUT	1A to 1Y 2A to 2Y 3A to 3Y 4A to 4Y	7	70	
10	t_{PHL_1} t_{PHL_2} t_{PLH_1} t_{PLH_2}		64 to 67 68 to 71 72 to 75 76 to 79	Same tests and terminal conditions as for subgroup 9, except T_A = 125° C.															7 7 7 7	165 80 180 110	
11	Same tests, terminal conditions and limits as for subgroup 10, except T_A = -55° C.																				

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4.5 Methods of examination and test. Methods of examination and test shall be as specified in the appropriate tables and as follows:

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional current and positive when flowing into the referenced terminal.

4.5.2 Life test cooldown procedure. When devices are measured at 25° C following application of the operating life or burn-in test condition, they shall be cooled to room temperature prior to removal of the bias. Alternately, the bias may be removed during cooling if the case temperature is reduced to room temperature within 30 minutes after removal of the test condition.

4.6 Inspection of preparation for delivery. Inspection of preparation for delivery shall be in accordance with MIL-M-38510, except that the rough handling test shall not apply.

5. PREPARATION FOR DELIVERY

5.1 Preservation-packaging and packing. Microcircuits shall be prepared for delivery in accordance with MIL-M-38510.

6. NOTES

6.1 Notes. The notes specified in MIL-M-38510 are applicable to this specification.

6.2 Intended use. Microcircuits conforming to this specification are intended for use for Government microcircuit applications (original equipment) and logistic purposes.

6.3 Ordering data. The contract or order should specify the following:

(a) Complete part number (see 1.2).
(b) Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.
(c) Requirement for certificate of compliance, if applicable.
(d) Requirements for notification of change of product or process to procuring activity in addition to notification to qualifying activity, if applicable.
(e) Requirements for packaging and packing, if other than level C of MIL-M-55565.
(f) Requirements for failure analysis (including required test condition of Method 5003), corrective action and reporting of results, if applicable.
(g) Requirements for product assurance options.
(h) Requirements for carriers, special lead lengths or lead forming, if applicable. These requirements shall not affect the part number.

6.4 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-STD-1313, MIL-STD-1331 and as follows:

GND - - - - - Electrical ground (common terminal)
V_{IN} - - - - - - Voltage level at an input terminal

6.5 Logistic support. Lead materials and finishes (see 3.3), are interchangeable. Unless otherwise specified, microcircuits procured for Government logistic support will be procured to device class "B" (see 1.2.2), lead material and finish "C" (see 3.3), and case outline "A" (see 1.2.3). There is no substitute for case outline "C". Longer length leads and lead forming shall not affect the part number.

6.6 Substitutability. Microcircuits covered by this specification are substitutable for commercial device type 54L86.

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Custodian:
Air Force - 17

Review activities:
Air Force - 11, 17, 80

User activity:
Air Force - 19

Preparing activity:
Air Force - 17

Agent:
DSA - ES

(Project 5962-F051)

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